Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/08/2010
PI Name:	Locke, James M.D.		
Project Title:	Survey of On-Orbit Sleep Quality (Sleep Quality Questionnaire)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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COI Name (Institution):			
Grant/Contract No.:	Directed Research		
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Performance Goal Text:			
	The NASA/JSC Behavioral Health and Performance Element of Space Medicine Division, Human Research Program (BHP) is collaborating with the Crew Health and Safety, Space Medicine Division, Space Operations Mission Directorate to recommend strategies and policies that will help astronauts experience improved sleep quality during spaceflight. The study provides an opportunity to extensively query current astronauts regarding their sleep behaviors and sleep quality on Earth, during training periods, and during Shuttle missions, utilizing a secure online survey as well as a follow-up interview. Sleep and circadian issues have been noted by astronauts since Apollo missions. Empirical research evaluating sleep quantity during spaceflight reveals some crew members are attaining reduced hours of sleep in-flight, at times even more pronounced prior to conducting critical mission tasks. Aspects of the spaceflight environment, such as loud noise and sleep restraint use, may contribute to sleep loss and fatigue. Given the large body of ground evidence that demonstrates sleep loss and fatigue negatively affect performance, it is essential to fully understand the individual, mission related, and		

Task Description:	environmental factors that affect sleep, and develop strategies and policies to mitigate the effects of these factors. The study serves as a revised, full version of the 2008 protocol, Sleep Quality Questionnaire Pilot Study. The pilot study consisted of interviewing ten astronauts returning from two STS missions regarding their sleep experience on earth and in space, as well as their opinions related to countermeasure effectiveness and recommendations for acquiring optimal sleep in-flight. The instrument used for conducting the interview consisted of 44 open-ended items. Analyses from the pilot study reveal differences in the sleep experience across the crews. As an example, eight of the ten astronauts indicated they used sleep medications in flight; seven of these astronauts reported no signs of side effects, while one astronaut indicated feeling specific side effects as a result of sleep medication use. Additionally, half of the respondents revealed that scheduling interfered with sleeping and discussed specific aspects of the scheduling that posed a problem for them in-flight. Preliminary findings from the pilot study have led to the refinement of the Sleep Quality Questionnaire. Primarily, the current efforts aim to gather data in a matter that will yield quantifiable, measurable results, so that relationships between different risk factors and the subjective sleep experience can be explored. The majority of the questionnaire includes Likert-scale items and was administered via a secure online portal. Analysis from the study will include data collected during the pilot study and will provide needed information for aiding the individual astronaut to experience improved sleep during spaceflight missions. Such recommended strategies and policies will lead to better management of sleep and fatigue by addressing crew health and habitability standards. Furthermore, it is anticipated that the environmental factors and mission operations of lunar sorties and the longer term lunar habitat missions will continue to pose challenges to the sleep and circadian system. As a result, findings from the proposed study will help inform and establish flight rules and requirements for future exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impact of this project is that it broadens understanding of the technical aspects of sleep in microgravity. The knowledge gained will further insight into what promotes/detracts from sleep quantity/quality in Space. Ultimately, the goal of this research is to identify measures to improve Astronauts' sleep. This research is unlikely to have significant terrestrial benefits.
Task Progress:	Significant progress has been made. Data collection is complete. Eligible enrollees in the study included all astronauts who have flown on the 16 Shuttle missions since STS-107. Excluded were individuals who were ferried to/from the ISS for long-duration missions. All of the eligible subjects were interviewed, with the exception of 4 individuals who declined participation. A total of 68 astronauts completed on-line surveys and were interviewed. All interviews have been transcribed. Analysis of the data reveals individual variability in the subjective sleep experience in space. Fifty four percent of shuttle crew members report difficulty falling asleep during their missions, while 43% report that sleep on orbit was interrupted. There is a significant correlation between assessments of falling asleep and staying asleep in space indicating that, generally, those who report difficulty falling asleep in space also reported difficulty staying asleep in space. Difficulty falling asleep on earth was significantly associated with difficulty staying asleep in space. Other factors positively associated with disturbed sleep include thinking about upcoming tasks, and sleep medication use. Sleep medications are more likely to be used during spaceflight missions than on earth, even when in “unfamiliar places” and in earth situations where sleep shifting is necessary. Interview responses revealed the importance of having time for wind-down and the methods through which crew members attained 'wind-down' in flight; strategies to ensure medications are more effectively used on orbit; recommendations for attaining comfortable sleep positions, including strapping, tension, or free floating; and, importantly, the individual differences in identifying effective sleep countermeasures.
Bibliography Type:	Description: (Last Updated:)